Unit-1

1. $P(A) + P(\bar{A}) = 1$

2. $P(A) = \dfrac{\text{no. of favourable cases}}{\text{Total no. of events}}$

3. $P(A) = 1 \Rightarrow$ certain event
   $P(A) = 0 \Rightarrow$ impossible event

$$nC_n = \frac{n!}{(n-r)!\,r!}$$

2mark: Axiom of Probability (or) Conditions of probability (or) Properties of probability

i) $0 \le P(A) \le 1$
ii) $P(S) = 1$
iii) $P(\cup A_n) = \sum_{n=1}^{\infty} P(A_n)$

Discrete random variable (Probability mass function)

$$P(X = x) = P_i$$
$$\Sigma P_i = 1$$

Continuous random variable (Probability density function)

$$f(X=x) = f(x)$$

i) $\int_{-\infty}^{\infty} f(x)\,dx = 1$ $\qquad \int_a^b f(x)\,dx = 1$
when $a \le x \le b$

Cumulative distribution function (Distributive function)

$$F(x) = P(X = x)$$
$$F'(x) = f(x)$$

Mean:

$$E(x) = \Sigma x P(x=x) = \Sigma x P_i$$

Variance

$$Var(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \Sigma x^2 P(x=x) \quad (or) \quad E(x^2) = E[x(x+1)] - E(x)$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Expected value: $E(x) = \bar{x} = \begin{cases} \Sigma x P(x), & x \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x)\,dx, & x \text{ is continuous} \end{cases}$

Note:

i) $E(a) = a$

ii) $E(ax) = aE(x)$

iii) $E(ax+b) = aE(x) + b$

iv) $E(x+y) = E(x) + E(y)$ [Addition theorem]

v) $Var(x) = E(x^2) - (E(x))^2$

vi) $var(ax+b) = a^2 var(x)$

n^{th} moment for discrete random variable:

$$E[x^n] = \sum_i x_i^n P_i = \mu_n' \quad , n \geq 1$$

n^{th} moment for continuous random variable

$$E[x^n] = \int_{-\infty}^{\infty} x^n f(x)\,dx = \mu_n'$$

n^{th} central moment for discrete

$$E[(x-\bar{x})^n] = \sum_i (x_i-\bar{x})^n P_i = \mu_n$$

n^{th} central moment for continuous

$$E[(x-\bar{x})^n] = \int_{-\infty}^{\infty} (x-\bar{x})^n f(x)\,dx = \mu_n$$

Relation b/w moments about origin and moments about mean:

i.) $\mu_1 = 0$

ii.) $\mu_2 = \mu_2' - (\mu_1')^2$

iii.) $\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$

iv.) $\mu_4 = \mu_4' - 4\mu_3'\mu_1 + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

Moment generating function

$$M_x(t) = E[e^{tx}] = \begin{cases} \sum_x e^{tx} p_x ; & \text{discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx ; & \text{continuous} \end{cases}$$

$$M_x(t) = E[e^{tx}] = E\left[1 + \frac{tx}{1!} + \frac{t^2x^2}{2!} + \dots \frac{t^n x^n}{n!} + \dots\right]$$

$$= 1 + tE(x) + \frac{t^2}{2!}E(x^2) + \dots + \frac{t^n}{n!}E(x^n) + \dots$$

$$M_x(t) = 1 + t\mu_1' + \frac{t^2}{2!}\mu_2' + \frac{t^3}{3!}\mu_3' + \dots + \frac{t^n}{n!}\mu_n'$$

μ_1' – coefficient of t in expansion of $M_x(t)$

μ_2' – " of $\frac{t^2}{2!}$ " " " "

μ_3' – " of $\frac{t^3}{3!}$ " " " "

$\mu_n' =$ coefficient of $\frac{t^n}{n!}$ in expansion of $M_X(t)$

$$\mu_n' = \frac{d^n}{dt^n}\left[M_X(t)\right]_{t=0}$$

$\mu_1' =$ Mean of x

$\mu_2' =$ variance of x

$$\int_0^\infty x^{n-1} e^{-x}\, dx = (n-1)!$$

$S.D = \sqrt{Var(x)}$

$$\int_0^\infty x^{n} e^{-x}\, dx = n!$$

Binomial distribution:

$P(x) = P(X=x) = nC_x\, p^x q^{n-x}$; $x = 0, 1, 2, 3 \ldots$

Mean $= np$

variance $= npq$

MGF $= M_X(t) = (q + pe^t)^n$

Poisson distribution:

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad ; \quad x = 0, 1, 2, 3 \ldots$$

Mean $= \lambda$

variance $= \lambda$

MGF $= e^{\lambda(t-1)}$

Geometric distribution:

$$P(X=x) = q^{x-1} p \; ; \quad x = 1, 2, 3$$

Uniform distribution

$$f(x) = \begin{cases} \frac{1}{b-a} , & a \le x, b \\ 0 , & \text{otherwise} \end{cases}$$

Mean: $E(x) = \mu_1' = \frac{b+a}{2}$

variance : $\mu_2 = \mu_2' = (\mu_1')^2 = \frac{1}{12}(b-a)^2$

M.G.F $\quad \mu_x(t) = \frac{1}{t(b-a)} \left[e^{tb} - e^{ta} \right]$

Exponential distribution:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} , & x \ge 0 \\ 0 , & \text{otherwise} \end{cases}$$

Mean $= \frac{1}{\lambda}$

variance $= \frac{1}{\lambda^2}$

M.G.F $= \frac{\lambda}{\lambda - t}$

Memory Less property of Exponential distribution:

$$P(x > s+t / x > s) = P(x > t)$$

Normal Distribution:

$$f(x;\sigma,\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$x \sim N(\mu, \sigma^2)$

If $x \sim N(\mu, \sigma^2)$ then

$$z = \frac{x-\mu}{\sigma}$$

Unit - 2

Joint probability mass function: (Discrete)

$$\sum_i \sum_j P(x_i, y_j) = 1$$

Joint probability density function: (continuous)

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$$

Joint cumulative distribution function:

i) Discrete:

$$F_{XY}(x, y) = \sum_i \sum_j P_{ij}$$

ii) Continuous:

$$F_{XY}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(x, y)\, dy\, dx$$

Marginal Probability density function:

i) Discrete:

$$P_X(x_i) = \sum_{j=1}^{\infty} P(x_i, y_j) \quad , i = 1, 2, 3 \ldots$$

$$P_Y(y_j) = \sum_{i=1}^{\infty} P(x_i, y_j), j = 1, 2, 3, \ldots$$

ii) Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx$$

Conditional Probability distribution function:

i) Discrete:

$$P_x\left(x_i/y_j\right) = \frac{P(X = x_i, Y = y_j)}{P(Y = y_j)} = \frac{P(x_i, y_j)}{P_y(y_j)}$$

$$P_y\left(y_j/x_i\right) = \frac{P(X = x_i, Y = y_j)}{P(X = x_i)} = \frac{P(x_i, y_j)}{P_x(x_i)}$$

ii) continuous:

$$f_y(y/x) = \frac{f_{xy}(x, y)}{f_x(x)}$$

$$f_x(x/y) = \frac{f_{xy}(x, y)}{f_y(y)}$$

two variables x and Y are independent

$$P(x_i, y_j) = P_x(x_i) \cdot P_y(y_j)$$

Ex: $x_i = 0, y_j = 0$

$$P(0,0) = P_x(0) \cdot P_y(0)$$

(or)

$$f(x,y) = f(x) \cdot f(y)$$

Correlation coefficient:

$$r_{xy} = \frac{E\{[x-E(x)]\,[y-E(y)]\}}{\sqrt{E\{x-E(x)\}^2 - E\{y-E(y)\}^2}}$$

$$r_{xy} = \frac{cov(x,y)}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$r_{xy} = \frac{E(xy) - E(x)E(y)}{\sqrt{[E(x)^2 - E^2(x)][E(y)^2 - E^2(y)]}}$$

$$r_{xy} = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2 [n\Sigma y^2 - (\Sigma y)^2]}}$$

$$cov(x,y) = E(xy) - E(x)E(y)$$

$$var(x) = E(x^2) - (E(x))^2$$

$$var(y) = E(y^2) - (E(y))^2$$

If A and B are independent

$$P(A \cap B) = P(A)\,P(B)$$

Note:

- $r_{xy} = 0$, when $cov(x,y) = 0$
  x and y are uncorrelated
- $r_{xy} \neq 0$, when $cov(x,y) \neq 0$
  x and y are correlated

Note:

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\bar{y} = \frac{\Sigma y}{n}$$

$$var(x) = \sigma_x^2 = \frac{1}{n}\Sigma u^2 - \left(\frac{\Sigma u}{n}\right)^2$$

$$var(y) = \sigma_y^2 = \frac{1}{n}\Sigma v^2 - \left(\frac{\Sigma v}{n}\right)^2$$

$$cov(x,y) = c_{xy} = \frac{1}{n}\Sigma uv - \left(\frac{\Sigma u}{n}\right)\left(\frac{\Sigma v}{n}\right)$$

Regression line of y on x:

$$y-\bar{y} = \frac{c_{xy}}{\sigma_x^2}(x-\bar{x})$$

(or)

$$y-\bar{y} = \frac{r_{xy}\sigma_y}{\sigma_x}(x-\bar{x})$$

Regression line of x on y:

$$x-\bar{x} = \frac{c_{xy}}{\sigma_y^2}(y-\bar{y})$$

(or)

$$x-\bar{x} = \frac{r_{xy}\sigma_x}{\sigma_y}(y-\bar{y})$$

$\because u = x-\bar{x}$, $v = y-\bar{y}$

Transformation of Random variables:

1) One dimensional Random variable:

g(x) is strictly increasing

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

2) Two dimensional random variable:

To find Jacobian $J\left(\frac{x, y}{u, v}\right) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$

$$f(u, v) = |J| \, f(x, y)$$

if v is not given $\therefore$ $v = y$

Unit - 3

Stationary process

i) $E(X(t))$ is constant

ii) $E(X^2(t))$ is constant

iii) $Var(X(t))$ is constant

iv) $R(t_1, t_2) = E\{X(t_1)X(t_2)\}$ is function of $(t_1 - t_2)$

WSS process:

i) $E(X(t))$ is constant

ii) $R(t_1, t_2) = E(X(t_1)X(t_2))$ is ~~constant~~ function of $(t_1 - t_2)$

SSS process

i) $E(X(t))$ is constant

ii) $Var(X(t))$ is constant

iii) $R(t_1, t_2) = E(X(t_1)X(t_2))$ is function of $(t_1 - t_2)$

$E(x(t)) = \Sigma x(t) P(x(t)=n) = \int x(t) f(\theta) d\theta$

$E(x^2(t)) = \Sigma x^2(t) P(x(t)=n)$

(or)

$\Sigma x(x+1) P(x(t)=n) - \Sigma x P(x(t)=n)$

Binomial

$(1-x)^{-2} = 1+2x+3x^2+4x^3+\ldots$

$(1-x)^{-3} = 1+3x+6x^2+10x^3+15x^4+\ldots$

$\sin(4\pi+\theta) = \sin(2\pi+\theta) = \sin\theta$

$\cos a \cos b = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$

$\cos a \sin b = \frac{1}{2}[\sin(a+b) - \sin(a-b)]$

$\cos(a+b) = \cos a \cos b - \sin a \sin b$

$\cos(a-b) = \cos a \cos b + \sin a \sin b$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin\theta \cos\theta = \cos\theta \sin\theta = \frac{\sin 2\theta}{2}$

Poisson process:

$P_n(t) = P(x(t)=n) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, \quad n=0,1,2\ldots$

mean $= \lambda t$

variance $= \lambda t$

if given $\lambda, t, P \Rightarrow P_n(t) = \frac{e^{-\lambda P t}(\lambda P t)^n}{n!}, \quad n=0,1,2\ldots$

Autocorrelation:

$$R_{xx}(\tau) = E[x(t)\, x(t+\tau)]$$
$$= E[x(t+\tau)\, x(t)]$$
$$= E[x(t)\, x(t-\tau)]$$
$$= E[x(t-\tau)\, x(t)]$$

$x(t)$ is stationary in SSS (or) WSS $E[x(t)\, x(t+\tau)]$ is function of τ. denoted by $R_{xx}(t)$, $R_x(t)$ or $R(\tau)$

Crosscorrelation:

$$R_{xy}(\tau) = E[x(t)\, y(t+\tau)]$$

$x(t)$ & $y(t)$ are WSS then $E[x(t)\, y(t+\tau)]$ is a function of τ denoted by $R_{xy}(t)$

If A and B are independent then $E[A] = E[B]$

Random telegraph:

α	-1	1
$P(\alpha)$	$1/2$	$1/2$

$$x(t) = \alpha(-1)^{N(t)}$$

Semi-random telegraph:

$$x(t) = (-1)^{N(t)}$$

Unit - 4

Power spectral density

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-i\omega\tau} d\tau$$

Wiener - Khichine theorem:

$$S(\omega) = \lim_{T \to \infty} \left[\frac{1}{2T} E\left\{ |X_T(\omega)|^2 \right\} \right]$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\int_{-\infty}^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_{0}^{\infty} x^n e^{-x} dx = n!$$

$$\int_{0}^{\infty} e^{at} \cos bt \, dt = \frac{e^{at}}{a^2+b^2} (a\cos bt + b\sin bt)$$

If $F[f(t)] = F(\omega)$ then

i.) $F\left[e^{-a|t|}\right] = \frac{2a}{e+\omega^2}$

ii.) $F\left[\cos(\omega_0 t)\right] = \pi\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]$

iii.) $F\left[\sin(\omega_0 t)\right] = \frac{\pi}{i}\left[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)\right]$

iv.) $R_{xx}(\tau) = F^{-1}\left[S_{xx}(\omega)\right]$

v.) $S_{xx}(\omega) = F\left[R_{xy}(\tau)\right]$

vi.) $R_{xx}(\tau - 2a) = e^{-i\omega 2a} = e^{-i2a\omega}$

vii.) $R_{xx}(\tau + 2a) = e^{i\omega 2a} = e^{i2a\omega}$

$$\int u\,dv = uv - u'v_1 + u''v_2 - \ldots$$

autocorrelation:

$$R(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, e^{i\tau\omega}\, d\omega$$

$$1-\cos 2\theta = 2\sin^2\theta$$
$$1-\cos\theta = 2\sin^2\frac{\theta}{2}$$

$$\cos\omega_0 t = \frac{e^{i\omega_0 t} + e^{-i\omega_0 \tau}}{2}$$

$$R_{xx}(t) = e^{-a\tau^2} \longrightarrow S_{xx}(\omega) = \sqrt{\frac{\pi}{a}}\, e^{-\frac{\omega^2}{4a}}$$

$$F^{-1}\left[\frac{1}{a^2+\omega^2}\right] = \frac{1}{2a} e^{-a|\tau|}$$

$$F^{-1}\left[\frac{1}{(1+\omega^2)^2}\right] = \frac{1}{4}(1+\tau)\, e^{-\tau}$$

$$\boxed{\text{Mean square value}} = E\left[X^2(t)\right] = R_{xx}(0) \text{ (or) } \boxed{\text{Average value}}$$

Cross spectral density

$$S_{xy}(\omega) = \int_{-\infty}^{\infty} R_{xy}(\tau)\, e^{-i\omega\tau}\, d\tau$$

$$S_{yx}(\omega) = \int_{-\infty}^{\infty} R_{yx}(\tau)\, e^{-i\omega\tau}\, d\tau$$

Properties:

i) $S_{xy}(\omega) = S_{xy}(-\omega)$

ii) $S_{xy}(\omega) = 0$, if $x(t)$ are orthogonal

iii) If $x(t)$ and $y(t)$ are uncorrelated
$S_{xy}(\omega) = E(X)E(Y)\,\delta(\omega)$

$$\mu_x^2 = \lim_{T\to\infty} R_{xx}(\tau)$$

$$E(x) = \text{Mean} = \mu_x = \sqrt{\mu_x^2}$$

$$E(x^2) = R_{xx}(0)$$

$$Var(x) = E(x^2) - [E(x)]^2$$

Linear system:

$$f\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 f\{x_1(t)\} + a_2 f\{x_2(t)\}$$

Time invariant:

If $f(x(t)) = y(t)$ then

$$f(x(t-\tau)) = y(t-\tau), \quad -\infty < \tau < \infty$$

$$F[R_{xx}(\tau)] = S_{xx}(\omega)$$

$$F[R_{xx}(\tau+2a)] = e^{i2a\omega}$$

$$F[R_{xx}(\tau-2a)] = e^{-i2a\omega}$$

$$y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du$$

Unit-5

sample size - n
population size - N
sample mean - $\bar{x}$
sample variance - s^2
Population mean - μ
population variance - σ^2

critical values (z_α) of z:

	Level of significance		
	1% (0.01)	5% (0.05)	10% (0.1)
Two tailed test	2.58	1.96	1.645
Right tailed test	2.33	1.645	1.28
Left tailed test	-2.33	-1.645	-1.28

Large samples:
Test of significance of single mean:

$$z = \frac{\bar{x}-\mu}{(\sigma/\sqrt{n})} \qquad z = \frac{\bar{x}-\mu}{(S/\sqrt{n})}$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n}\sum_{i=1}^{1} (x_i-\bar{x})^2$$

$$v(\bar{x}) = \frac{\sigma^2}{n}$$

$$E[s^2] = \frac{n-1}{n}\sigma^2$$

$$E(s^2) = \sigma^2, \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i-\bar{x})^2$$

Test of significance for difference of mean:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

note: $\sigma_1 = \sigma_2 = \sigma$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} \cdot \frac{1}{n_2}}}$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1 > \bar{x}_2$ (left tail)

$\bar{x}_1 < \bar{x}_2$ (right tail)

Test of significance of single proportion

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad [P + Q = 1]$$

$Q = 1 - P$

Intervals

$P = p \pm 3\sqrt{\frac{pq}{n}}$

$q = 1 - p$

Test of significance of difference of proportions

$$Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

t-Test:

type-I

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$\bar{x} \Rightarrow$ mean $= \frac{\Sigma x}{n}$

$$s^2 = \frac{1}{n-1} \Sigma (x-\bar{x})^2$$

(or)

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$\bar{x} = \frac{\Sigma x}{n}$

$s^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$

s^2 - sample variance

σ^2 - population variance

μ - population mean

$\bar{x}$ - sample mean

degrees of freedom $= n-1$

Intervals : $\bar{x} + t_{\%} \left(\frac{s}{\sqrt{n-1}}\right)$

Type-II

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

degrees of freedom $= n_1 + n_2 - 2$

F-Test:

$$F = \frac{s_1^2}{s_2^2} = \frac{n_1 s_1^2 (n_2-1)}{(n_1-1)\, n_2 s_2^2}$$

$$s_1^2 = \frac{\Sigma (x_1 - \bar{x_1})^2}{n_1 - 1}$$

$$\boxed{\bar{x_1} = \frac{\Sigma x_1}{n} \qquad \bar{x_2} = \frac{\Sigma x_2}{n}}$$

$$s_2^2 = \frac{\Sigma (x_2 - \bar{x_2})^2}{n_2 - 1}$$

if $s_1^2 > s_2^2$

$$F = \frac{s_1^2}{s_2^2}$$

if $s_1^2 < s_2^2$

$$F = \frac{s_2^2}{s_1^2}$$

degrees of freedom $= (n_1 - 1, n_2 - 1)$

χ^2 - test:

$$\chi^2 = \frac{\sum_{i=1}^{n} (O_i - E_i)^2}{E_i} \qquad dof = n-1$$

For 2×2 contigency table

$$\psi^2 = \frac{(a+b+c+d)(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)}$$

$$dof = (2-1)(2-1) = 1$$